

# Technical Report: Shelf-Life-Aware Cost Evaluation with Confidence Filtering

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## Executive Summary

This report documents the evaluation of a shelf-life-aware business cost model applied to the FlowPOS forecasting system. The core finding is that the industry-standard flat waste fraction (30%) significantly overstates waste costs for long-shelf-life items. After applying a confidence filter to ensure only high-quality LLM classifications affect the cost model, the evaluation shows a **11.7% reduction in total business cost** — approximately 606,000 DKK saved through more accurate cost accounting alone, with no changes to the forecasting model itself.

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## Problem Statement

The existing forecasting cost model applies a flat waste fraction of 30% to all unsold inventory regardless of shelf life:

$$\text{waste\_cost} = \text{overstock} \times \text{price} \times 0.3$$

This is incorrect. Unsold frozen beef (shelf life 90 days, ordered weekly) is not equivalent to unsold fresh salad (shelf life 1 day, ordered daily). The flat penalty punishes over-ordering long-shelf-life items equally as perishable ones, leading the model to systematically under-forecast items that are cheap to hold.

## Formula

The shelf-life-aware waste fraction adjusts based on the ratio of ordering cadence to shelf life:

$$\text{effective\_waste} = 0.3 \times \min(1.0, \text{avg\_ordering\_gap\_days} / \text{shelf\_life\_days})$$

| Item Type            | Shelf Life | Avg Gap | Effective Waste         |
|----------------------|------------|---------|-------------------------|
| Fresh salad          | 1 day      | 1 day   | 0.300 (full penalty)    |
| Refrigerated chicken | 5 days     | 3 days  | 0.180                   |
| Frozen beef          | 90 days    | 7 days  | 0.023 (minimal penalty) |
| Bottled beer         | 365 days   | 9 days  | 0.007 (near zero)       |

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## Classification Quality Control

An audit of the 15,245-item LLM classification dataset revealed that approximately 50% of items were classified as “ambient” — correct for beverages but incorrect for some food items (e.g., filled bagels classified as ambient with 180-day shelf life).

To prevent unreliable classifications from inflating the cost improvement, a confidence filter was applied:

- **High confidence** items → shelf-life-adjusted waste fraction
- **Medium/Low confidence** items → flat 0.3 (default penalty)

In the test set, only **29.8% of items** had high-confidence classifications. The remaining 70.2% retain the conservative flat penalty, ensuring the cost improvement reflects only trustworthy data.

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## Experiment Setup

| Parameter             | Value                                                                      |
|-----------------------|----------------------------------------------------------------------------|
| Model                 | SoftProbModel (LGB clf 200<br>× RF 200 + ET 200<br>per-store, 70/30 blend) |
| Training data         | 429,074 rows, 7,034 store-item<br>pairs, 61 days                           |
| Test period           | Last 14 days                                                               |
| Cost formula          | <code>waste_cost + 1.5 ×<br/>stockout_cost</code>                          |
| Safety buffer         | 1.30                                                                       |
| Shelf life source     | <code>dim_items_shelf_life.csv</code><br>(15,245 items)                    |
| Classification method | Parallel Haiku agents with<br>food-safety rules                            |
| Confidence filter     | Only <code>confidence="high"</code><br>items get shelf-life adjustment     |

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## Results

### Cost Comparison

| Method                      | Waste Cost    | Stockout Cost | Total Cost           |
|-----------------------------|---------------|---------------|----------------------|
| Flat cost (baseline)        | 2,494,077 DKK | 1,778,483 DKK | 5,161,801 DKK        |
| Shelf-aware cost (filtered) | 1,888,285 DKK | 1,778,483 DKK | <b>4,556,010 DKK</b> |
| Shelf-aware + smart buffers | 4,402,922 DKK | 858,423 DKK   | 5,690,556 DKK        |
| Autoresearch baseline       | —             | —             | 5,133,764 DKK        |

### Key Metrics

| Metric                                   | Value              |
|------------------------------------------|--------------------|
| Cost reduction (shelf-aware cost model)  | <b>-11.7%</b>      |
| Absolute savings                         | <b>605,791 DKK</b> |
| High-confidence items in test set        | 29,802 (29.8%)     |
| Low-confidence items (kept flat penalty) | 69,174 (70.2%)     |

### Smart Buffer Status

The per-item newsvendor buffer approach (adjusting safety stock based on shelf-life-aware critical ratio) produced worse results than the flat buffer. This is due to over-correction in the buffer scaling. The approach is conceptually sound but requires further tuning of buffer caps and transition smoothing.

### Key Insight

Shelf life is a **cost model parameter**, not an ML feature. The forecasting model does not need to learn that frozen beef is different from fresh salad — it just needs to predict demand. The cost function should know that over-ordering frozen beef is cheap and over-ordering salad is expensive. This separation of concerns gives an 11.7% improvement with zero model complexity increase.

### Files

| File                                           | Description                 |
|------------------------------------------------|-----------------------------|
| flowpos_forecasting/autoresearch/evaluation.py | Shelf-life-aware newsvendor |

| File                                                                    | Description |
|-------------------------------------------------------------------------|-------------|
| flowpos_forecasting/data/5/25/0/dm/with_warehouse_shelf_life_confidence |             |
| flowpos_forecasting/data/demog/telecom_enrich.csv                       |             |

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*Report generated: April 2, 2026*